AnyCam 3.9.0-rc2.0 Audit Report

Release audit - 25 October 2026 - 2 items - tag 3.9.0-rc2.0

A release candidate with the scope CrystalHeeler confirmed on 2026-10-25: B84 and B85. The first field night of 3.9.0-rc1.0 (24-25 Oct, Test System B) recorded 29 headlight files, 9 of the motion-sensitive light by the hot tub and 2 of the dawn switch, although the headlight pause started at the right moment (docs/Recording_Review_2026-10-24.md, sections 9 and 10).

What the field night showed

Cause	Files	Fix
The pause ended (4 s, 6 s in all) while the light still crossed the yard; the slow-movement check had kept lit pictures from the pause	20	B84
A zone passed on the light's faint first edge, before the pause saw it	4	B85
One lit picture recorded at once (the one-picture rule)	3	B85
No pause at all	2	B84 (longer pause), B85
The dawn infrared-to-colour switch under the 75% light rule	2	B85
The motion-sensitive light switching on or off	9	B85
Insects in the infrared, several pictures in a row	10	Not in scope (B86)

The changes

Item	Change
B84	A light pause lasts up to 8 s (12 s in all); a one-sided change within 3 s after a pause continues it with no quiet picture before it; the slow-movement check keeps no picture during a pause or for 0.6 s after; a detector restart clears the pause (the log had shown "passing light over after -284.0 s").
B85	A live start waits 1.25 s (MOTION_HOLD_S); the pre-roll grows by the wait. It is cancelled by a light pause, a light change, the camera switching between infrared and colour (colour jump 1.0 or more, one side 2.5 or less; every picture's colour is now kept), or a change 90% one-sided that stopped at once (two still comparisons). A colour switch also counts as a light change outside the wait. The one-picture rule ignores a lit lawn, and its hit counts twice, so the log no longer says "one picture only" for a picture it recorded.
Buffer	MOTION_BUF_KEEP_S = 3 + 12 + 1 + 1.25 = 17.25 s (was 10 s): about 17 MB at 8 Mbit/s per armed camera; MOTION_BUF_MAX_S stays 20 s.

Evidence: the replay

409 recordings (every first file of 20-24 Oct on the patio camera, and the whole night of 24-25 Oct) were decoded once to the detector's pictures (4 a second, 128x96 or 64x48 grey, each picture's colour) and fed through the add-on's own _motion_feed, with each camera's zones and sensitivity from the settings copy of 24 Oct 22:42. Starts and stops were caught where the add-on hands them to asyncio.

Night of 24-25 Oct	3.9.0-rc1.0	3.9.0-rc2.0
Headlights (29 files)	22	0
Motion-sensitive light (8)	4	0
Dawn switch (2)	1	0
A person (3)	3	3
Insects (9)	9	9

All 409: by day 214 of 215 still record (the one lost is a dusk headlight); at night 36 of the 127 that 3.9.0-rc1.0 records. All 91 lost were viewed: headlights, the porch light, the dusk and dawn switches, an insect. The cat, the person at the door, both door openings and the moth still record. Pause lengths 4/6 s, 6/10 s and 8/12 s left 2, 1 and 0 of the 29 headlight files; 8/12 s lost no other file.

Tests run

Suite	Result	Scope
Release gate	**PASS**	Eight gates; 63 contracts, new ones for _motion_begin, _motion_hold_check, _motion_colour, _motion_colour_switch and _motion_slow_look; the feed and detector contracts updated.
Python integration	**PASS**	812 of 812; new section AV (14 checks); M1 updated for the 17.25 s buffer.
JavaScript behaviour	**PASS**	217 of 217; no page change.
Recording replay	**PASS**	409 recordings, whole motion path, old against new.
Field	**NOTED**	Not run yet: an rc for field testing.

Not tested

- **The live detector.** The replay decoded files with OpenCV; the add-on decodes the live sub-stream with ffmpeg. Each file starts 3-8 s before its trigger, so the replay misses what came before.
- **The Pi's memory** with the 17.25 s buffer on several armed cameras.
- **Dusk on 3.9.0-rc1.0 or rc2.0.** 3.9.0-rc1.0 started after dark on 24 Oct; the dusk switch was replayed from 20-24 Oct files only.
- **The rabbit** that switched the light on was not found in the 9 lamp files; an animal that keeps moving records in the tests (AV2).

Best-practices compliance

Standard	Check applied	Result	Note
Rule 2	Logs before code	**PASS**	From the field log of 25 Oct and its 54 videos.

Standard	Check applied	Result	Note
Rule 8	Regression first in our code	**PASS**	The causes were in AnyCam: pause length, slow-check references, the one-picture rule.
P1 Python	No duplicate definitions; small functions	**PASS**	The wait is four functions; gate 1 clean.
P4 Async	No blocking I/O on the loop	**PASS**	The colour of each picture is a sum over 6,144 bytes; no I/O added.
P17 Testing	Tests before the change ships	**PASS**	Section AV; the replay of 409 recordings.
Rule 9 (privacy)	New text	**PASS**	Cameras named by role; no addresses.

Files changed

File	Change
anycam_motion.py	B84, B85: the wait, the colour switch, the pause, the slow check, the one-picture rule, the 17.25 s buffer.
camera_discovery.py, config.yaml	3.9.0-rc2.0.
verify_release.py, tests/test_server.py	Contracts; section AV; M1.
CHANGELOG.md, docs	3.9.0-rc2.0; build plan; recording review sections 9 and 10.